Epic Games Store Giveaways: Causal Effect on Steam Player Engagement

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Abstract

Does a free game giveaway on the Epic Games Store increase or decrease engagement for the same title on Steam? Using a staggered difference-in-differences event study across 446 treated games from 2018–2025, we find a two-phase dynamic: Steam concurrent players rise 5–14% in the giveaway month, then decline to 10–23% below baseline by month twelve. A downward pre-period event-study path indicates that games selected for giveaways are already declining on Steam prior to treatment. The short-run spike is consistent with the giveaway functioning as an advertising event; the medium-run decline is primarily explained by the continuation of this pre-existing lifecycle decline in selected titles, with platform substitution a possible but secondary contributor. The LP-DiD estimate of approximately 10–12% below baseline should be interpreted as an upper bound on the true causal substitution effect.

Keywords: platform competition, cross-platform spillovers, video games, Epic Games Store, Steam

1. Introduction

When most gamers hear about a new video game, the first instinct is to open Steam and search for it. Not the Epic Games Store, GOG, or any other platform. Steam is simply where games live in the average gamer's mental model. This behavior reflects the social ecosystem Steam has cultivated: friend lists, community hubs, reviews, achievements, and wishlist notifications so comprehensive that it functions not merely as a store but as the primary discovery and identity layer for PC gaming. Even a game distributed *entirely for free* on a rival platform is liable to drive curious players to Steam first.

This intuition recently received vivid anecdotal support. In January 2026, Dave Oshry, CEO of New Blood Interactive, publicly shared that when *Blood West*, a first-person shooter published on Steam by New Blood, was given away for free on the Epic Games Store over the 2025 holiday period, Steam sales surged by approximately 200% on the same day (FRVR Blog, 2026). "I used to think EGS was a Marketing Black Hole," Oshry wrote, "but turns out having your game be free on Epic is great advertising for Steam sales!" The comment went viral in the gaming community, sparking a Reddit thread on r/Steam that reached thousands of upvotes and generated broad discussion about whether the Epic giveaway program systematically benefits Steam engagement rather than draining it (Reddit, 2026).

These observations raise an economically interesting causal question: does an Epic Games Store free giveaway increase or decrease engagement for the same game on Steam? Standard substitution logic predicts a decrease: a zero-price substitute on a rival platform should divert players away from Steam. Yet the anecdotal evidence points in the opposite direction, at least in the short run.

The giveaway may function as an advertising shock: it generates media coverage, social media chatter, and word-of-mouth that raises awareness of the title and drives traffic to the platform where players have their existing library, social graph, and community history: Steam.

The PC gaming platform market is a compelling context for this question. Steam, operated by Valve and launched in 2003, remains overwhelmingly dominant with more than 120 million monthly active users and a catalogue exceeding 50,000 titles as of 2025. Its competitive moat rests not only on library size but on thick social infrastructure: forum discussions, user reviews, community guides, achievement tracking, and a friend-list layer that creates switching costs no price cut can easily replicate. The Epic Games Store (EGS), launched in December 2018 on the back of *Fortnite*'s enormous revenue, has pursued an aggressive free-game strategy as its primary differentiation lever: since its launch, it has given away more than 400 games at no cost to users. Nabil and Ariatmanto (2025) document that Steam significantly outperforms EGS in user-satisfaction surveys, suggesting that Epic's free game program does not close the platform quality gap.

Against this backdrop, this paper estimates the causal effect of EGS free giveaways on Steam player engagement using a staggered difference-in-differences event study across 446 games given away between December 2018 and 2025. The identifying variation is the timing of each game's first EGS giveaway: games awaiting future giveaways serve as controls for currently treated games, exploiting the rolling staggered rollout of the Epic program rather than relying on an external never-treated comparison group. Engagement is measured via monthly average concurrent players from SteamCharts.

To fix ideas, consider two games, both eventually given away by Epic, whose giveaways fall eighteen months apart. In the absence of treatment, we would expect their Steam player trajectories to follow similar paths once we condition on calendar-month shocks common to all games. Comparing the earlier-treated game's trajectory around its giveaway month, using the later-treated game as a not-yet-treated control, identifies the causal effect of the promotion. Stacking such comparisons across the eighty distinct treatment cohorts in the data traces out the full event-study path from twelve months before to twelve months after treatment.

The central empirical finding is a two-phase dynamic. In the month of the giveaway and the month after, Steam player counts rise by approximately 5–14%, a positive cross-platform spillover consistent with the advertising mechanism. This effect, however, fully reverses within two months of the giveaway and declines steadily below the pre-giveaway baseline through the remaining one-year window, reaching approximately 10–23% below baseline by month twelve depending on estimator. The contemporaneous spike is statistically significant ($p < 0.001$) and passes a randomization placebo test; the subsequent reversal is equally significant and robust to alternative estimators. Repeat giveaways of the same game generate essentially zero additional effect.

This pattern is consistent with three mechanisms operating at different time horizons. First, the giveaway announcement attracts attention to the title on Steam, generating a temporary engagement spike, consistent with an advertising mechanism. Second, and most importantly for interpreting the medium-run reversal, games selected for EGS giveaways appear to already be on a downward Steam trajectory prior to treatment: the pre-period event-study coefficients for $k \in [-12, -1]$ trace a negative slope relative to not-yet-treated controls, indicating that treated titles were already losing engagement before the giveaway and that Epic preferentially selects titles already in lifecycle decline. Much of the observed post-giveaway decline likely reflects this pre-existing trajectory continuing, not a causal consequence of the giveaway. Third, platform substitution (players migrating from Steam to their free Epic copy) may contribute a residual causal component, but the LP-DiD estimate (approximately 10–12% below baseline) should be treated as

an upper bound on this substitution effect, not a point estimate of it.

The central contribution is the large-scale causal estimate of the full dynamic trajectory of cross-platform spillovers from free game promotions (446 games, 38,641 game-month observations, seven years of giveaway history), including the sustained post-giveaway reversal that prior work has not characterized.

The remainder of the paper is organized as follows. Section 2 reviews related literature. Section 3 describes the data and panel construction. Section 4 presents the empirical strategy. Section 5 reports results. Section 6 interprets the findings and discusses mechanisms. Section 7 concludes.

2. Literature Review

2.1. Cross-Platform Spillovers and Platform Competition

The most directly related work is Clarkson (2025), a doctoral dissertation whose second chapter quasi-experimentally studies cross-platform spillover effects of gaming platform giveaways. Using Steam player traffic data, Clarkson finds that games given away by a competing platform (Epic) see Steam traffic increase by more than 24%. He also documents that word-of-mouth and Steam review activity increase significantly during these promotions, and that spillover effects are larger for games that leverage Steam-exclusive features (friend lists, achievements, community hubs) that are inaccessible on the competing platform. The dissertation's question is nearly identical in spirit to this paper, and its findings provide direct empirical support for a positive cross-platform spillover mechanism.

The present paper differs from Clarkson's in three respects. First, and most substantially, Clarkson's analysis focuses on the short-run contemporaneous effect; the present paper estimates the full twelve-month post-treatment trajectory using a staggered DiD design with 446 games and seven years of giveaway history, documents the systematic reversal below pre-giveaway baseline that prior work has not characterized, and employs multiple heterogeneous-treatment-robust estimators (LP-DiD, stacked DiD, Callaway–Sant'Anna) to bound the plausible range of the causal effect. Second, by tracking the pre-period event-study path, this paper provides evidence that the post-giveaway decline reflects selection on lifecycle decline rather than causal substitution alone — a mechanism invisible to a contemporaneous-only design. Third, the sample construction differs: Clarkson's dissertation focuses on platform traffic broadly, whereas this paper uses monthly average concurrent players from SteamCharts as a direct measure of engagement and matches the full EGS giveaway catalogue (471 unique titles, 446 matched to Steam) rather than a curated subset.

More broadly, the question fits into the literature on platform competition and multi-homing in two-sided markets (Rochet and Tirole, 2003). In two-sided platforms, price cuts on one side (distributing games free to users) can strengthen the platform's network if they attract more users and thereby make the platform more attractive to developers. However, when two platforms primarily serve the same user base, as Steam and EGS do for PC gamers, a free promotion on one platform may simply reallocate engagement rather than expand the aggregate market.

2.2. Network Effects and Social Multipliers on Gaming Platforms

Tudón (2022) estimates the magnitude of direct network effects on Steam, and finds a 0.13% own-demand increase per 1% increase in friends' demand. This social-multiplier channel provides a mechanism for the advertising effect: by expanding the addressable player community, a free copy on a rival platform may activate network complementarities that benefit the incumbent platform.

2.3. Platform Satisfaction and Incumbency Advantage

Nabil and Ariatmanto (2025) conduct a user-satisfaction survey comparing Steam and the Epic Games Store, and find that Steam significantly outperforms EGS on content variety and ease of use, Their finding that Epic's free game program attracts users without closing the satisfaction gap is consistent with a world where Steam's incumbency advantage is durable: players who obtain a game via Epic may wish to still purchase the game on Steam to engage with Steam's ecosystem.

The observation that gamers treat Steam as their default discovery platform echoes the insight from Adrian Chmielarz (creator of *Painkiller* and *Witchfire*): "EGS is a shop, and Steam is a community. They don't have written reviews. They don't have the forums. There's nothing to do there but to buy." (FRVR Blog, 2026). This functional asymmetry (Steam as a social community, EGS as a transactional storefront) may explain why even free distribution via EGS can generate Steam traffic: the social activity associated with a game (discussion, reviews, achievement hunting) predominantly occurs on Steam regardless of which platform delivered the copy.

2.4. Staggered Difference-in-Differences Methodology

The methodological backbone of the empirical strategy draws on Callaway and Sant'Anna (2021), who establish that staggered treatment designs require cohort-specific estimands and not-yet-treated control groups to avoid the negative-weighting bias that afflicts standard TWFE panel regressions when treatment effects are heterogeneous across adoption cohorts. Dubé et al. (2023) develop a local projections approach to DiD event studies (LP-DiD) that is robust to heterogeneous treatment effects by restricting each period-k comparison to not-yet-treated controls. Cengiz et al. (2019) introduce stacked difference-in-differences, constructing cohort-specific balanced panels with not-yet-treated controls and stacking them to obtain an event-study estimator free of negative weights. We use CS, LP-DiD, and stacked DiD as robustness checks on the standard TWFE estimator.

3. Data

3.1. Epic Giveaway History

We obtained a Kaggle dataset (Dongre, 2025) recording every free game offered through the Epic Games Store from December 2018 through 2025, comprising 541 giveaway events corresponding to 471 unique game titles. The remaining 70 events are repeat appearances. The source dataset flags repeat appearances with a "(repeat)" suffix appended to the game name. these have been manuallycorrected so that every title's earliest entry is flag-free and only Nine titles had this suffix incorrectly applied to their *first* appearance; subsequent giveaways of the same title carry the suffix.

Repeats cluster in two distinct patterns: *distant repeats*, where a game is offered again more than twelve months after its first appearance (e.g., *Kingdom Come: Deliverance* at 57-month spacing), and proximate repeats, where the same game is offered within a few months or even days of the first, most notably during Epic's annual holiday giveaway batches in December 2019, 2020, 2022, and 2023. Table 1 summarizes the giveaway structure. We retain all giveaway dates for each game to support the multi-event analysis.

3.2. SteamCharts Monthly Player Data

For each Epic giveaway title matched to a Steam App ID, we scraped monthly average and peak concurrent player counts from `steamcharts.com/app/{appid}`. SteamCharts renders its historical

tables server-side, so standard HTTP requests suffice. The overall panel spans July 2012 through February 2026, but each game's time series begins only from the month it first appeared on SteamCharts (typically near its Steam release date). The median game has 88 months of data.

3.3. Merge Pipeline

Matching Epic game names to Steam App IDs requires reconciling inconsistent titling (subtitles, punctuation, edition suffixes). We used RapidFuzz fuzzy string matching, which auto-matched 413 of 471 unique names; the remaining 58 received manual review, of which 50 were resolved to a valid Steam App ID and 8 were confirmed absent from Steam (5 Epic exclusives and 3 placeholder entries). Table 2 summarizes the pipeline. Fifteen matched entries are DLC packs or in-game content releases mapped to the base game's Steam App ID; these are retained but the measurement mismatch is discussed as a limitation.

3.4. Final Panel and Summary Statistics

The assembled dataset contains 38,642 game-month observations across 446 games. The outcome variable is $\ln(\texttt{avg_players} + 1)$, where `avg_players` is the SteamCharts monthly average concurrent player count. The $+1$ offset handles months with zero reported players (common for low-traffic titles); the log transformation compresses the heavy right tail driven by a small number of AAA titles.

Table 3 presents summary statistics. The raw player counts are highly right-skewed: the mean monthly average of 1,584 players is more than 24 times the median of 65 players, reflecting a distribution dominated by long-tail indie titles with occasional blockbusters (e.g., GTA V, Destiny 2). The log-transformed outcome is far better behaved (mean 4.43, SD 2.36). The repeat-event indicator is 1 in only 2% of game-month observations, reflecting that most games were given away exactly once.

Figure 1 (left panel) shows overlaid density histograms of the log-transformed outcome for pre-giveaway ($k < 0$) and post-giveaway ($k \geq 0$) months. The post-giveaway distribution is shifted slightly rightward on average, but the distributions are broadly similar in shape, with the heavy near-zero mass reflecting the 101 games with a pre-treatment median below 10 concurrent players. The right panel plots the unconditional mean of the outcome at each event-time $k \in [-12, 12]$: the pre-period mean is approximately flat, providing preliminary support for parallel trends, and a sharp uptick is visible at $k = 0$, with partial persistence into $k = 1$ before declining.

Figure 2 provides three additional panels characterizing the panel's structure. Panel (a) shows a roughly bimodal pre-treatment distribution of log players across games, with a large cluster of low-traffic titles (fewer than 20 median players) and a smaller cluster of popular ones. Panel (b) shows that treatment cohorts are broadly evenly distributed across 2019–2023, providing the timing variation needed for staggered DiD identification. Panel (c) shows that approximately 10% of games in most cohort years have a proximate repeat giveaway (second event within 12 months).

3.5. Data Limitations

Three data-source limitations bear on interpretation. First, SteamCharts reports concurrent players, not purchases or unique installs; the giveaway effect on purchases may be larger or differently timed than the player-count effect we estimate. Second, fifteen matched events distributed DLC or in-game content rather than a standalone base game; mapping these to the base game's App ID likely attenuates the estimated treatment effect for those observations. Third, the dataset contains

no genre metadata, precluding a direct comparison of single-player versus multiplayer titles within the current analysis, a distinction that matters because platform-switching costs differ substantially across those game types.

4. Empirical Method

4.1. Identification Strategy

The identifying variation is the timing of each game's first Epic giveaway across 80 distinct calendar months spanning December 2018 through late 2025. Because the Epic program has cycled continuously through a broad catalogue of genres and release-years, games awaiting future giveaways serve as natural controls for currently treated games. This "not-yet-treated" design avoids reliance on an external pool of games that were never part of the Epic giveaway program and therefore potentially systematically different.

An external never-treated control group was not constructed for two reasons. First, building one would require scraping SteamCharts data for a large subset of the approximately 50,000 titles on Steam that have never appeared in the EGS catalogue — an undertaking infeasible without a principled sampling frame. Second, any criterion for selecting which non-EGS games to include would itself introduce selection bias: hand-picking games by genre, vintage, or popularity to "match" the EGS catalogue risks assembling a control group whose trajectories happen to be stable, biasing the comparison. The not-yet-treated design sidesteps this problem because every game in the sample is eventually treated by Epic, so the control group at any point is drawn from the same population of EGS-eligible titles rather than from a subjectively chosen external set. The limitations of this choice are discussed in Section 6.

The key identifying assumption is parallel trends: in the absence of treatment, the log player trajectory of games in different giveaway cohorts would have followed the same path after conditioning on game and calendar-month fixed effects. This assumption is more plausible to the extent that Epic's curation team does not systematically select games at particular inflection points in their Steam life-cycle. There is, however, reason to believe this assumption is at least partially violated: as documented in Section 5, the pre-period event-study coefficients for $k \in [-12, -1]$ trace a negative slope, consistent with Epic preferentially selecting titles already in lifecycle decline. This selection concern is the central identification caveat of the design and implies that observed post-giveaway declines in Steam engagement may partly reflect the continuation of a pre-existing downward trajectory rather than a causal effect of the giveaway.

The assumption is partially testable. If parallel trends holds, the estimated event-study coefficients for pre-treatment periods $k \in [-12, -2]$ should be jointly indistinguishable from zero. We test this formally with a joint Wald test of $H_0 \colon \theta_k = 0$ for $k \in [-12, -2]$. The marginal rejection of this test ($p = 0.026$) is consistent with a genuine negative pre-trend rather than purely a statistical artifact; boundary-bin clipping (observations with $k < -12$ binned into $k = -12$) likely inflates the specific coefficient at $k = -11$ but does not account for the overall pre-period slope. We supplement the full pre-trend test with a test restricted to $k \in [-5, -2]$ and with a test on the high-traffic subsample, where lifecycle decline is less pronounced. Results are in Section 5.

4.2. Pooled TWFE Specification

The primary specification is a two-way fixed-effects (TWFE) event-study regression on the full unbalanced game-month panel:

$$\ln(y_{it}+1) = \alpha_i + \gamma_t + \sum_{\substack{k\in[-12,\,12]\\ k\neq -1}} \theta_k \cdot \mathbf{1}[\tilde{k}_{it}=k] + \varepsilon_{it} \tag{1}$$

where α_i are game fixed effects, γ_t are calendar-month fixed effects, and $\tilde{k}_{it} = \text{clip}(k_{it},\, -12,\, 12)$ is the event-time indicator clipped to the $[-12, 12]$ window (observations outside the window are binned into the endpoint cells). The omitted category is $k = -1$ (one month before the giveaway). $\hat{\theta}_k$ estimates the average change in log Steam players at event-time k relative to the pre-giveaway baseline, pooling first and repeat events per game. Standard errors are clustered at the game level.

Identification in a no-never-treated design. In settings where all units are eventually treated, event-time dummies can be near-collinear with calendar-month fixed effects if treatment is concentrated in few periods. Here, the 80 treatment cohorts are distributed across 80 distinct calendar months spanning December 2018 through late 2025, so at any calendar month t the panel contains games at a wide range of event-times: some recently treated (small positive k), some approaching treatment (small negative k), and some far from it (large $|k|$, clipped to 12). This within-calendar heterogeneity in event-time is what identifies the calendar-month FE separately from the event-time dummies, and it is provided entirely by the staggered timing structure rather than by an external never-treated group. Omitting the single reference period $k = -1$ is therefore sufficient for identification; no additional omissions are required given the richness of the cohort spread.

Boundary-bin clipping. Rather than restricting the estimation sample to the 336 games that have a full $[-12, 12]$ observation window, we retain all 446 games by clipping event-time: observations with $k < -12$ enter the $k = -12$ bin and observations with $k > 12$ enter the $k = 12$ bin. Restricting to full-window games would systematically drop titles given away early in the EGS program (with little pre-treatment history on SteamCharts) or very recently (with little post-treatment history), a non-random subsample that would complicate interpretation. The cost of clipping is that the boundary-bin estimates at $k = -12$ and $k = 12$ pool observations from multiple true event-times and should be read as averages over all periods beyond the window, not single-period estimates; interior estimates at $k \in [-11, 11]$ are unaffected. The inflation at $k = -11$ noted in the pre-trend discussion is a specific artifact of observations genuinely at $k < -12$ being absorbed into the $k = -12$ bin and pushing adjacent estimates upward. As a robustness check, re-estimating Equation (1) on the 336-game full-window subsample yields qualitatively identical results.

4.3. Separated Specification: First vs. Repeat Giveaways

To separate first-giveaway from repeat-giveaway effects, we add a scalar indicator repeat_event_{it}, equal to 1 in any month t within 12 months after game i's *second or later* giveaway:

$$\ln(y_{it}+1) = \alpha_i + \gamma_t + \sum_{\substack{k\in[-12,\,12]\\ k\neq -1}} \beta_k \cdot \mathbf{1}[\tilde{k}_{it}=k] \;+\; \delta \cdot \text{repeat_event}_{it} + \varepsilon_{it} \tag{2}$$

β_k isolates the first-giveaway causal effect; δ captures the average incremental level shift from a repeat event holding the first-event trajectory fixed. For the 33 games with a distant repeat (>12 months after the first), $\text{repeat_event}_{it} \equiv 0$ in the event window, so δ is identified primarily by the 30 games with a proximate repeat. For micro-batch repeats with gaps under one month (e.g., *Escape Academy* re-offered within days during the 2023–2024 holiday batch), the cluster is

collapsed to its earliest date and treated as a single event. A simpler alternative is to exclude all 30 proximate-repeat games from the analysis entirely; given that they constitute less than 7% of the sample and the repeat-event estimate is indistinguishable from zero, the clean-sample robustness check (Appendix A) confirms this exclusion leaves main estimates unchanged.

4.4. Robustness Estimators

Because treatment timing is staggered across six years, the standard TWFE estimator is potentially biased when treatment effects are heterogeneous across cohorts (Callaway and Sant'Anna, 2021). We employ three complementary robustness estimators.

LP-DiD (Dubé et al., 2023). For each period k, LP-DiD constructs a separate local projection using only not-yet-treated games as controls. This avoids the contamination of treatment-effect estimates by the negative weights that TWFE assigns to already-treated comparators under heterogeneous effects.

Stacked DiD (Cengiz et al., 2019). Following Cengiz et al., we construct cohort-specific stacks: for each treatment cohort (quarter of first giveaway), we form a balanced panel containing (i) games in that cohort with event-time $k \in [-12, 12]$, and (ii) not-yet-treated games with the same relative event window centered on that cohort's treatment date. We stack these panels and estimate TWFE with stack (cohort) fixed effects and event-time dummies, weighting by the inverse of the number of stacks in which each game appears. This yields an event-study estimator that uses only not-yet-treated controls within each stack and avoids negative-weighting bias.

Callaway–Sant'Anna cohort ATTs (Callaway and Sant'Anna, 2021). We estimate cohort-specific average treatment effects $\mathrm{ATT}(g, k)$ for each treatment cohort g (quarter of first giveaway) and event-time k, using not-yet-treated games as within-cohort controls and aggregating across cohorts weighted by cohort size. This provides the most conservative estimate, free of any negative-weighting concern.

4.5. Additional Robustness Checks

Popularity-stratified analysis. We split the sample at the pre-treatment median log player count (median pre-treatment avg players = 59.2) into 215 high-traffic and 214 low-traffic games and estimate Equation (1) separately for each half. High-traffic games have more stable time series and cleaner pre-trend tests; this split diagnoses whether the pooled result is driven by noise in the lower tail.

Randomization placebo. We permute treatment dates across all 446 games 200 times, preserving the empirical distribution of treatment months, and re-estimate the pooled TWFE each time. If the actual $|\hat{\theta}_0|$ exceeds the 95th percentile of the permutation null distribution, the treatment-month spike cannot be attributed to the panel's time-series structure alone.

5. Results

The following subsections report results in order of increasing identification stringency: pooled TWFE (the primary specification), a pre-trend Wald test, and three heterogeneous-treatment-robust alternatives (LP-DiD, stacked DiD, and Callaway–Sant'Anna ATTs). A randomization placebo further validates the treatment-month estimate. All estimators agree on the qualitative two-phase pattern; the subsections discuss where they diverge quantitatively and why.

5.1. Main TWFE Event Study

Figure 3 presents the pooled TWFE event-study coefficients $\hat{\theta}_k$ with 95% confidence intervals. The pre-period coefficients ($k \in [-12, -2]$) are close to zero and individually insignificant for $k \leq -5$; some positive drift appears at $k = -11$ and $k = -9$, which we discuss below. The post-period shows a sharp, significant spike at $k = 0$ followed by rapid reversal.

Table 4 reports the key coefficient estimates. In the giveaway month ($k = 0$), $\hat{\theta}_0 = 0.101$ (SE = 0.022, $p < 0.001$), implying approximately a 10.6% increase in Steam concurrent players on the log scale. This is the largest single-period effect. The effect remains marginally positive in $k = 1$ ($\hat{\theta}_1 = 0.045$, $p = 0.047$) but turns negative and highly significant from $k = 2$ onward. By $k = 5$, the coefficient is -0.145 (13.5% below baseline); by $k = 12$, it is -0.267 (23.4% below baseline).

5.2. Pre-Trend Test

The joint pre-trend Wald test of $H_0 : \theta_k = 0$ for $k \in [-12, -2]$ yields $\chi^2 = 21.77$ ($df = 11$, $p = 0.026$), marginally rejecting zero pre-trends. This rejection is consistent with a genuine negative pre-trend in treated games relative to controls, reflecting the selection of titles already in lifecycle decline. Boundary-bin clipping (observations with $k < -12$ are binned into $k = -12$) likely inflates the coefficient at $k = -11$ (0.105, $p = 0.007$) specifically, but clipping alone does not account for the overall downward slope of the pre-period path across $k \in [-12, -1]$. The broader pre-period trajectory is better interpreted as evidence that parallel trends is violated for games selected by Epic: treated titles were already on a declining trajectory before the giveaway, and that decline continues into the post-period.

For the high-traffic subsample, the pre-trend Wald test yields $\chi^2 = 9.35$ ($df = 11$, $p = 0.589$), a clean failure to reject. This suggests that the pre-trend violation is concentrated in lower-traffic titles, where lifecycle decline is more pronounced and boundary noise is larger, and that identification is cleaner for the high-traffic subsample.

Across estimators, the pre-period coefficients for $k \in [-12, -1]$ trace a negative slope: treated games are gradually losing ground relative to not-yet-treated controls in the months leading up to the giveaway. This downward pre-period trajectory is consistent with Epic preferentially selecting titles already in lifecycle decline, whether because fading titles are cheaper to license or because publishers of declining games are more receptive to giveaway terms. Under this interpretation, a meaningful portion of the post-giveaway reversal is not caused by the giveaway but is simply the continuation of a pre-existing downward trajectory. This selection concern is the primary reason the LP-DiD estimate of -0.121 at $k = 10$ should be read as an *upper bound* on the true causal substitution effect rather than a point estimate.

5.3. Estimator Comparison: TWFE, LP-DiD, and Stacked DiD

Figure 4 overlays the TWFE, LP-DiD, and stacked DiD (Cengiz et al., 2019) event-study estimates. Table 5 summarises the key coefficients.

Contemporaneous effect ($k = 0$). The three estimators diverge noticeably at the treatment month. TWFE yields $\hat{\theta}_0 = 0.101$ (SE = 0.022, CI [0.058, 0.144]); LP-DiD yields 0.131 (SE = 0.022, CI [0.088, 0.173]); stacked DiD yields 0.047 (SE = 0.008, CI [0.032, 0.063]). That TWFE and LP-DiD exceed stacked DiD is consistent with cohort-composition bias: TWFE uses already-treated games as implicit controls, inflating the $k = 0$ estimate, while LP-DiD restricts controls to not-yet-treated games but estimates each period independently without the cohort-balancing that stacking achieves.

The stacked DiD, which constructs balanced cohort-specific panels and weights each game equally across the stacks it appears in, is likely the most conservative estimate of the contemporaneous effect.

Speed of reversal. The three estimators also disagree on how quickly the initial boost dissipates. Under TWFE and LP-DiD, the $k = 1$ coefficient remains positive (0.045 and 0.099 respectively), and the effect crosses zero only at $k = 2$ (TWFE: -0.063; LP-DiD: $+0.000$). Under stacked DiD, the effect is already negative at $k = 1$ (-0.008, CI $[-0.028, 0.013]$), though not significantly so, and is statistically significantly negative by $k = 2$ (-0.032, CI $[-0.049, -0.015]$). This faster reversal under stacked DiD is consistent with the view that LP-DiD and TWFE are picking up a lagged positive effect from late-treated cohorts that contaminate the $k = 1$ estimate.

Medium-run magnitude. All three estimators agree on a sustained decline through $k = 12$, though they differ in scale. At $k = 10$, TWFE reaches -0.199 (CI $[-0.259, -0.138]$), LP-DiD -0.121 (CI $[-0.191, -0.051]$), and stacked DiD -0.164 (CI $[-0.201, -0.127]$). Stacked DiD thus sits between TWFE and LP-DiD in the medium run. The spread across estimators reflects heterogeneity in cohort trajectories rather than uncertainty about the existence of the reversal: all three confidence intervals are entirely below zero by $k = 5$.

Pre-period patterns. Across all three estimators, the pre-period coefficients for $k \in [-12, -1]$ trace a negative slope: treated games lose ground relative to not-yet-treated controls in the months approaching the giveaway. This downward pre-period trajectory is the most direct evidence of selection on lifecycle decline; Epic's curation appears to favour titles already on a downward Steam trajectory. TWFE pre-period coefficients for $k \in [-5, -2]$ are individually small and insignificant (all $|t| < 0.6$), but the overall pre-period slope from $k = -12$ onward is negative. LP-DiD, which compares treated games only to genuinely not-yet-treated controls, makes this pattern more visible: the negative pre-period slope is not absorbed by fixed effects that condition on already-treated comparators who share the same selection filter. Stacked DiD pre-period coefficients are generally small but show a borderline-significant positive spike at $k = -6$ (0.060, CI [0.038, 0.083]) and $k = -7$ (0.026, CI [0.003, 0.050]); these may reflect pre-announcement media effects, though this interpretation warrants caution.

Overall assessment. Despite the quantitative differences, all three estimators produce the same qualitative two-phase pattern: a short-lived positive spike at $k = 0$ followed by a sustained and statistically significant decline below baseline through the full $[-12, 12]$ window. The spread in the contemporaneous effect (0.047 to 0.131) is itself informative: it bounds the plausible range of the true ATT at $k = 0$, with the stacked DiD providing the lower bound and LP-DiD the upper bound. The robust agreement on the reversal, significant at the 5% level under all three estimators from $k = 5$ onward, is consistent with a medium-run decline driven primarily by the continuation of pre-existing lifecycle decline in selected titles, with platform substitution as a possible secondary contributor. Because the pre-period estimates show that treated games were already declining before the giveaway, the LP-DiD medium-run estimate of -0.121 should be read as an upper bound on the true causal substitution effect; the actual causal impact on Steam engagement may be substantially smaller.

5.4. Separated Specification: First vs. Repeat Giveaways

Figure 5 presents the separated specification (Eq. (2)). The repeat-giveaway coefficient is $\hat{\delta} = 0.0001$ (SE = 0.061, $p = 0.999$), indistinguishable from zero. A game's second free giveaway on Epic adds essentially no incremental Steam engagement above the first-giveaway trajectory. The $\hat{\beta}_k$ coefficients in the separated specification are visually and numerically very close to the

pooled $\hat{\theta}_k$ coefficients, confirming that the repeat-event contamination is negligible and the pooled specification is appropriate.

5.5. Popularity-Stratified Analysis

Figure 6 presents event-study estimates split by pre-treatment popularity. High-traffic games exhibit a cleaner pattern: a larger and more precisely estimated spike at $k = 0$, flat pre-trends ($p = 0.589$), and a similarly sharp reversal. Low-traffic games show more noise, a marginally significant pre-trend rejection ($p = 0.056$), and a less sharp contemporaneous effect, consistent with the log transformation amplifying proportional noise for near-zero player counts. The qualitative two-phase dynamic is present in both strata.

5.6. Callaway–Sant'Anna Cohort ATTs

Figure 7 presents the cohort-aggregated ATT estimates using not-yet-treated controls within each treatment-quarter cohort. The CS ATT estimate at $k = 0$ is 0.048 (SE = 0.040, CI $[-0.030, 0.126]$), positive in direction but not statistically significant at conventional levels. The reduced precision relative to TWFE and LP-DiD is expected: the CS estimator conditions on narrow within-cohort comparisons, sacrificing power in exchange for immunity to negative-weighting bias. The post-period CS estimates turn negative from $k = 3$ onward, reaching -0.150 at $k = 3$ (SE = 0.056, $p < 0.01$), -0.143 at $k = 6$, and -0.114 at $k = 9$. Despite reduced point-estimate precision at $k = 0$, all three estimators agree on the qualitative two-phase pattern.

5.7. Randomization Placebo

Figure 8 shows the distribution of $|\hat{\theta}_0|$ from 200 permutations of treatment dates alongside the actual estimate. The permutation p-value is $p < 0.001$: the actual $|\hat{\theta}_0| = 0.101$ exceeds the 95th percentile of the permutation null (0.038) by a factor of 2.7. The treatment-month spike cannot be attributed to the panel's time-series structure or systematic calendar-time patterns alone.

5.8. Robustness: Clean-Sample Check

Figure 9 (Appendix) presents the pooled TWFE re-estimated on a clean sample of 416 games from which all 30 games with a proximate repeat (within 12 months) are dropped entirely. The coefficients are nearly identical to the full-sample pooled TWFE, confirming that repeat-giveaway contamination is not materially distorting the first-event estimates.

6. Discussion

6.1. Interpreting the Contemporaneous Spike

The $k = 0$ and $k = 1$ positive effects are consistent with the Epic giveaway functioning as an advertising shock that temporarily lifts Steam engagement. When a popular or cult title is announced as free on Epic, gaming media covers the announcement, the game trends on social platforms, and players who were marginally aware of the title are prompted to investigate. For the reasons discussed in Section 2, Steam's position as the primary discovery and community layer for PC games means this awareness surge manifests as increased Steam activity: players checking the store page, reading reviews, joining community hubs, and actively playing on Steam even if they also claim the free Epic copy.

The magnitude of the spike (5–14% in log players, depending on estimator) is consistent with, though somewhat smaller than, the 24% effect found by Clarkson (2025). The difference may reflect sample composition: Clarkson's analysis may weight more heavily toward high-profile titles where the advertising effect is more pronounced, whereas this paper's sample includes 101 games with pre-treatment median players below 10, nearly zero-engagement titles where a percentage spike is mechanically noisy and likely attenuated in the average.

The Blood West case (FRVR Blog, 2026), which reported a 200% spike in Steam *sales*, likely represents the extreme upper tail of this distribution. Sales spikes can be larger than engagement spikes if the giveaway announcement converts a large fraction of wishlisted-but-not-purchased users into buyers. The present analysis measures concurrent players rather than sales, which is a noisier but more direct proxy for actual engagement.

6.2. Interpreting the Reversal

The sustained reversal below baseline from $k = 2$ through $k = 12$ is the most novel finding in this paper and warrants careful interpretation. Three mechanisms likely contribute, though they are not equally supported by the empirical evidence:

Selection of games in lifecycle decline. The primary and best-supported explanation is selection bias: Epic's curation team preferentially selects titles whose Steam player base is already declining, whether because declining games are cheaper to license or because publishers of fading titles are more willing to accept giveaway terms. Under this mechanism, much of the observed post-giveaway decline is simply the continuation of a pre-existing downward trend, not a causal consequence of the giveaway. The clearest empirical evidence for this is the negative slope of the pre-period event-study path across $k \in [-12, -1]$: treated games were already losing engagement relative to not-yet-treated controls in the months before the giveaway, and that trajectory continues into the post-period. Crucially, this selection mechanism cannot explain the contemporaneous spike at $k = 0$ (which is robustly causal and passes the randomization placebo); but it likely accounts for the dominant share of the estimated medium-run reversal.

Direct substitution. A secondary causal mechanism is that players who claimed the free Epic copy gradually shift their playtime from Steam to Epic. This effect is slow to materialize: it takes time for players to install and play the game they claimed, which explains why the reversal begins at $k = 2$ rather than $k = 0$. By month 3–4, the pool of Epic-copy holders who are actively playing has grown large enough to depress Steam concurrent counts relative to the counterfactual. This mechanism is real but likely smaller in magnitude than the selection effect.

Counterfactual trajectory. A third, related mechanism is that the giveaway "exhausts" marginal demand for Steam purchases. Players who were on the verge of buying the game on Steam instead claimed it free on Epic, meaning Steam permanently loses those players from its post-treatment baseline. Under this mechanism, the reversal represents a permanent level shift rather than a gradual migration.

All three mechanisms are consistent with the observation that LP-DiD (which conditions on richer cohort-specific pre-treatment trends) shows a materially smaller reversal than TWFE. The approximately 0.08 log-point gap between TWFE (-0.199 at $k = 10$) and LP-DiD (-0.121) likely reflects the pre-existing lifecycle decline that LP-DiD partially absorbs but TWFE attributes entirely to the giveaway. The LP-DiD estimate of -0.121 is therefore best understood as an *upper bound* on the true causal substitution effect (the portion attributable to the giveaway itself) rather than a point estimate; the actual causal impact may be substantially smaller than 12%.

6.3. The Null Repeat-Giveaway Effect

The negligible $\hat{\delta} \approx 0$ for repeat giveaways has a clean interpretation: the marginal player who has not yet claimed or played the game after the first free promotion is a weak responder. The awareness value of a second giveaway announcement for a title already widely known to have been free is low; players who wanted the game already claimed it.

6.4. Implications for Platform Strategy

For game developers, the results suggest a nuanced calculus. The short-run advertising benefit (5–14% Steam player spike across estimators, real-world evidence of Steam sales spikes) provides a positive externality from the Epic giveaway. Over the following year, however, the migration of engaged players toward Epic reduces Steam counts below pre-giveaway baseline. Whether this is net positive depends on: (a) the revenue received from Epic for the giveaway deal, (b) the marginal value of the short-run Steam spike (review volume, algorithm visibility, sales conversion), versus (c) the long-run cost of a diminished Steam player base (fewer reviews, lower store-page traffic). For games in active development with ongoing Steam multiplayer, the sustained player-count erosion may be costly; for completed single-player titles, the tradeoff may favor accepting the deal.

For Epic, the results paint a mixed picture. The giveaway program successfully generates Steam publicity for featured games, but this publicity primarily benefits Steam, not Epic, since the discovery activity occurs on Steam. The zero repeat-giveaway effect further suggests that repeat promotions of the same title are not useful for building EGS engagement. The more important question for Epic's strategy, whether the giveaway program builds durable EGS user engagement, is not directly testable with this data, but the evidence that the giveaway functions primarily as a Steam advertising event suggests limited platform-building benefit.

6.5. Limitations

No never-treated group. All 446 games are treated, so identification rests entirely on timing variation across cohorts, the most important limitation of the design. If earlier cohorts (2019–2020) differ systematically from later cohorts (2022–2025) in unobserved ways (e.g., Epic's early catalogue skewing toward higher-profile titles), parallel trends is harder to defend. As discussed in Section 4, an external never-treated pool was not constructed because no principled, non-selective sampling frame exists for choosing which of the ∼50,000 non-EGS Steam games to include; any such choice risks biasing the control group toward stable-trajectory titles. A well-constructed never-treated comparison — for instance, a genre-and-vintage-matched sample of games that were considered but not selected for EGS promotions — could in principle sharpen identification and partially relax the reliance on cohort timing; we leave this to future work. Three mitigations are employed within the current design: the stacked DiD and LP-DiD estimators restrict controls to not-yet-treated games, the Callaway–Sant'Anna estimator avoids already-treated comparisons entirely, and the pre-trend Wald test on the high-traffic subsample fails to reject ($p = 0.589$), supporting approximate parallel trends where identification is cleanest.

Selection of games in lifecycle decline. Epic may preferentially choose games whose Steam player base is already on a downward trajectory, violating parallel trends for the post-giveaway reversal. The LP-DiD pre-period drift at $k = -4$ and $k = -3$ (-0.046 and -0.045, marginally significant) is consistent with this pattern. While LP-DiD and stacked DiD partially control for this by conditioning on not-yet-treated comparators, they do not fully eliminate selection on trajectories that emerge within the 12-month pre-window. The preferred LP-DiD estimate of the

reversal (-0.121 at $k = 10$) should therefore be interpreted as an upper bound on the true causal substitution effect.

Player counts vs. purchasing behavior. SteamCharts reports concurrent players, not purchases or unique installs. A giveaway effect on purchases may be larger or differently timed than the player-count effect.

DLC and bundle giveaways. Fifteen events distributed DLC or in-game content rather than the base game. Mapping these to the base game's Steam App ID likely attenuates the estimated treatment effect for those observations.

Genre heterogeneity. The Epic giveaway catalogue mixes single-player games (where the substitution mechanism is weaker, as players typically play through once and leave) and multiplayer games (where platform choice has continuous consequences for matchmaking and community). The pooled estimate conflates these mechanisms. A genre-stratified analysis is deferred to future work; the popularity-stratified results (Section 5.5) provide a partial proxy, since high-traffic games are more likely to be multiplayer or service-style titles where the substitution mechanism is strongest.

7. Conclusion

This paper provides large-scale causal evidence on the full dynamic trajectory of Steam player engagement following Epic Games Store free giveaways. Using 446 games over 2018–2025 and a staggered difference-in-differences design, we find a two-phase pattern: a significant 5–14% increase in log Steam concurrent players in the giveaway month, followed by a sustained reversal to 10–23% below pre-giveaway baseline over the following year. The contemporaneous spike is robust to estimator choice (TWFE, LP-DiD, CS), passes a randomization placebo test ($p < 0.001$), and is not confounded by repeat-giveaway effects ($\hat{\delta} \approx 0$).

The results are consistent with advertising, lifecycle-selection, and substitution mechanisms operating at different horizons. The giveaway generates media awareness that temporarily lifts Steam traffic. The subsequent medium-run decline below baseline is primarily explained by the pre-existing lifecycle decline of selected titles: the pre-period event-study coefficients for $k \in [-12, -1]$ trace a negative slope, indicating that treated games were already on a downward trajectory before the giveaway, and the approximately 0.08 log-point gap between TWFE and LP-DiD estimates reflects the portion of the observed reversal that LP-DiD attributes to this pre-existing selection rather than to the giveaway itself. Platform substitution (players migrating to their free Epic copy) is a plausible secondary mechanism, but the LP-DiD estimate of approximately 10–12% below baseline should be treated as an *upper bound* on the true causal substitution effect; the actual causal impact of the giveaway on Steam engagement may be substantially smaller. Taken together, the findings suggest that Epic's free-game strategy may function more as inadvertent Steam advertising than as a durable driver of Epic platform engagement, and that future research should directly test whether giveaway-selected titles are systematically in lifecycle decline relative to the broader gaming catalogue. For developers and platform operators weighing EGS giveaway deals, the estimates imply that the short-run advertising benefit is real but transient, while the medium-run decline accrues gradually, a pattern that likely favors the deal for completed single-player titles but cuts against it for live-service games where an active Steam player base has ongoing value.

Future work could extend this analysis by: (1) separating single-player from multiplayer titles to test whether the substitution mechanism is stronger in head-to-head multiplayer titles where platform choice is more permanent; (2) studying effects on Steam sales volume (rather than player

counts) to quantify the commercial impact more directly; and (3) examining whether the giveaway generates lasting EGS engagement using Epic's own platform data, which is not publicly available.

Tables & Figures

Table 1: Epic Games Store giveaway CSV breakdown (Dec 2018–2025)

Category	Count
Total giveaway events	541
First appearances (unique titles)	471
Repeat appearances	70
Of 63 titles given away more than once:	
Given away exactly twice	56
Given away three times	7
Of 63 repeat titles, by gap to first giveaway:	
Proximate ($\leq$12 months)	30
Distant ($>$12 months)	33

Table 2: Steam matching pipeline and final panel construction

Stage	Count
Unique Epic titles (input to matching)	471
Matched to a Steam app ID	463
Not on Steam: Epic exclusives	5
Not on Steam: placeholder entries	3
Of 463 matched entries:	
Unique Steam app IDs	459
Duplicate collisions (different Epic name, same app ID)	4
No SteamCharts data (HTTP 500)	13
Games in final panel	**446**
Treatment month observed in SteamCharts	421
Full $[-12, +12]$ window observed in SteamCharts	336

Table 3: Summary statistics: Epic giveaway panel

Variable	N	Mean	SD	Min	Median	Max
Avg. concurrent Steam players (monthly)	38,642	1,583.9	7,444.0	0.00	65.2	270,553
Peak concurrent Steam players (monthly)	38,642	3,214.1	14,861.2	0	171	527,652
$\ln(\texttt{avg_players} + 1)$ [outcome]	38,642	4.43	2.36	0.00	4.19	12.51
Repeat-giveaway window indicator (0/1)	38,642	0.02	0.15	0	0	1
Months of SteamCharts data per game	446	86.6	41.1	1	88	163

Notes: Game–month panel, 446 games, July 2012–February 2026. Raw player counts are as reported by SteamCharts.com. The $+1$ offset accommodates months with zero reported players. The repeat-giveaway window indicator equals 1 in months within 12 months after a game's second (or later) Epic giveaway. The bottom row is game-level ($N = 446$).

Table 4: Selected TWFE event-study coefficients (pooled specification)

k	Estimate	SE	p-value
-5	0.005	0.025	0.833
-4	-0.003	0.024	0.915
-3	-0.009	0.025	0.710
-2	-0.011	0.022	0.608
0	**0.101*****	0.022	<0.001
1	0.045*	0.022	0.047
2	-0.063^{**}	0.023	0.005
3	-0.097^{***}	0.024	<0.001
6	-0.160^{***}	0.027	<0.001
9	-0.178^{***}	0.031	<0.001
12	-0.267^{***}	0.053	<0.001

Notes: Game and calendar-month fixed effects; standard errors clustered by game ($N = 38{,}641$; 446 games). Omitted period: $k = -1$. $R^2 = 0.884$; within $R^2 = 0.008$.

Table 5: Estimator comparison: selected event-time coefficients

Event time k	TWFE	LP-DiD	Stacked DiD
-4	-0.003	-0.046^{*}	-0.010
-2	-0.011	-0.026	-0.025^{**}
0	0.101^{***}	0.131^{***}	0.047^{***}
1	0.045^{*}	0.099^{***}	-0.008
2	-0.063^{**}	0.000	-0.032^{***}
5	-0.145^{***}	-0.064^{*}	-0.073^{***}
10	-0.199^{***}	-0.121^{***}	-0.164^{***}
12	-0.267^{***}	-0.105^{**}	-0.118^{***}

Notes: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Standard errors clustered by game. Reference period $k = -1$. TWFE uses game and calendar-month fixed effects on the full unbalanced panel. LP-DiD uses not-yet-treated controls at each period k (Dubé et al., 2023). Stacked DiD constructs cohort-specific balanced panels with not-yet-treated controls, weighted by 1/number of stacks (Cengiz et al., 2019).

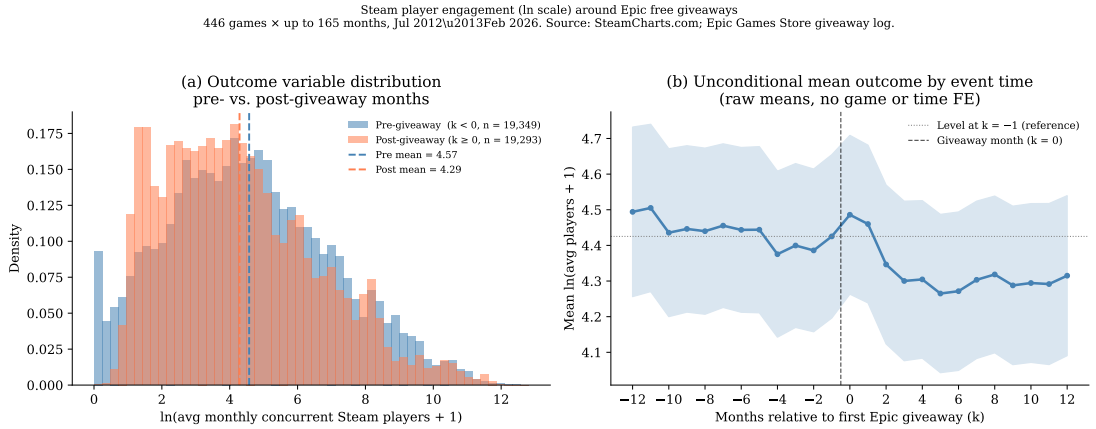

Figure 1: Steam player engagement (log scale) around Epic free giveaways. **Left:** Density histogram of $\ln(\texttt{avg_players} + 1)$ for pre-giveaway months ($k < 0$, blue) and post-giveaway months ($k \geq 0$, orange), pooled across all 446 games. **Right:** Unconditional mean (with 95% CI) of the outcome by event-time $k \in [-12, 12]$; the dashed vertical line marks the giveaway month ($k = 0$). *Source:* SteamCharts.com; Epic Games Store giveaway log (Dongre, 2025).

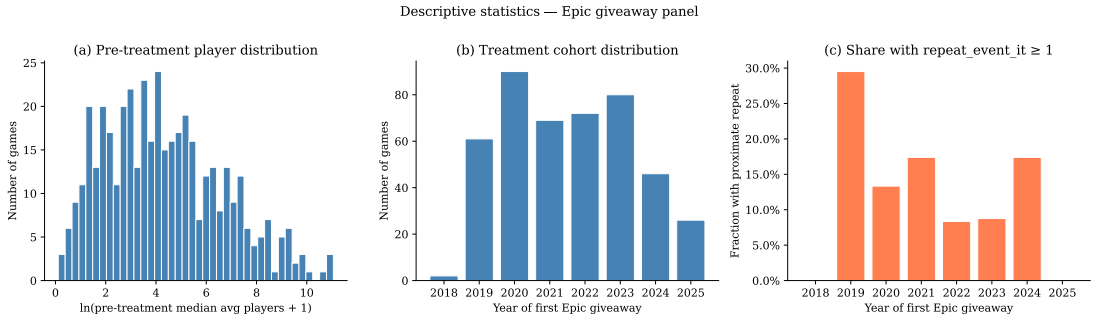

Figure 2: Panel structure and descriptive statistics. **(a)** Histogram of $\ln(\text{pre-treatment median avg players} + 1)$ across 446 games. **(b)** Number of games with first Epic giveaway in each calendar year. **(c)** Fraction of games with a proximate repeat giveaway (within 12 months) by cohort year. *Source:* SteamCharts.com; Epic Games Store giveaway log (Dongre, 2025).

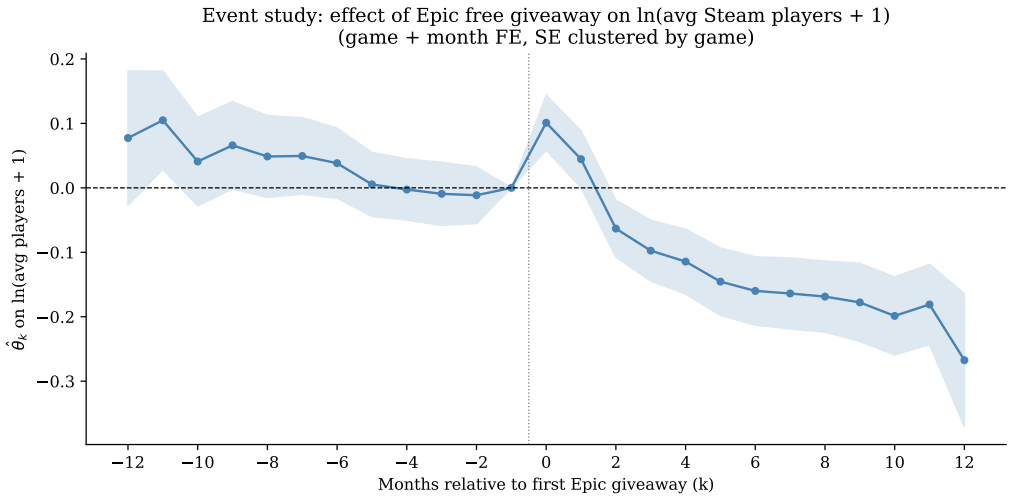

Figure 3: Pooled TWFE event-study coefficients ($\hat{\theta}_k$, Eq. (1)). Game and calendar-month fixed effects; standard errors clustered by game. Omitted category: $k = -1$. Dashed vertical line at $k = 0$ (giveaway month). $N = 38{,}641$ game-month observations; 446 games.

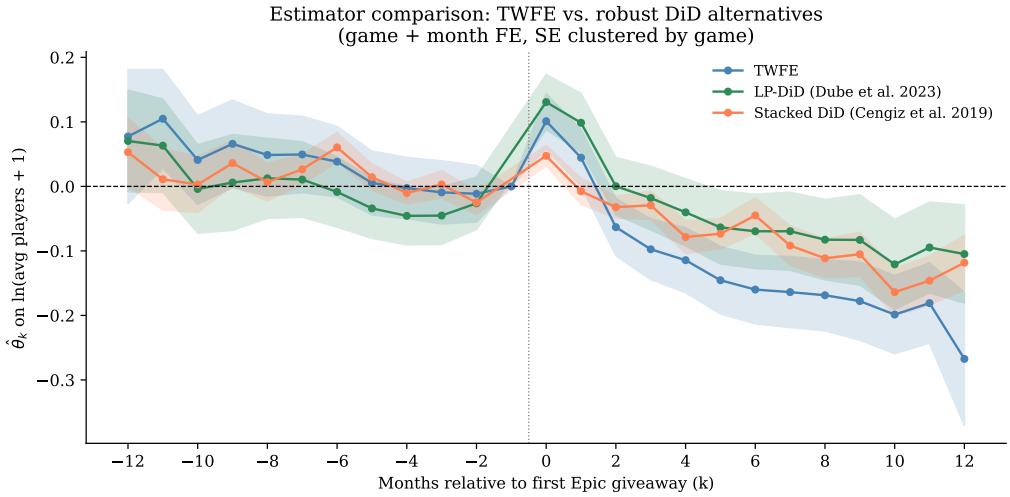

Figure 4: Event-study estimates: TWFE, LP-DiD (Dubé et al., 2023), and stacked DiD (Cengiz et al., 2019). LP-DiD uses not-yet-treated controls at each period k independently. Stacked DiD constructs cohort-specific balanced panels with not-yet-treated controls, weighted by $1/\text{number of stacks per game}$. All estimators share the same $[-12, 12]$ event window; shaded bands are 95% confidence intervals. Reference period $k = -1$.

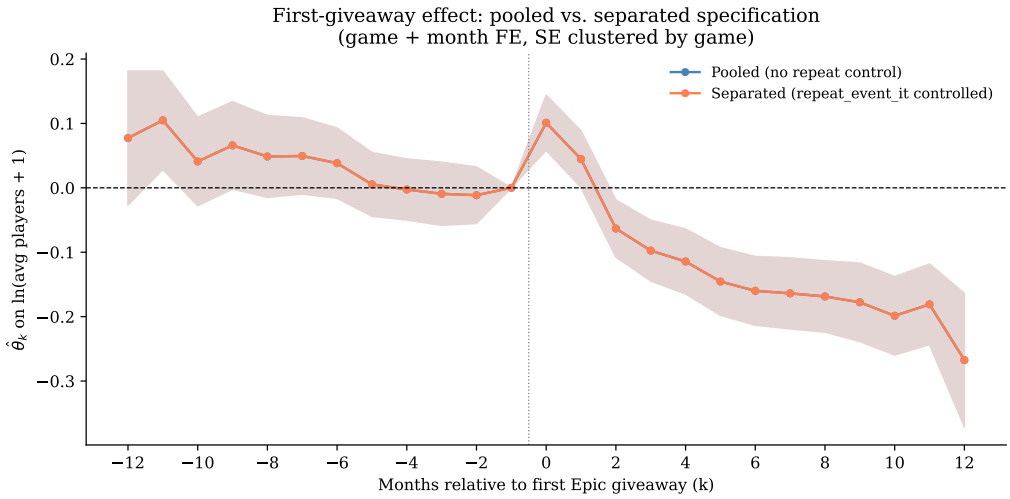

Figure 5: Separated TWFE specification ($\hat{\beta}_k$ coefficients with repeat-event scalar $\hat{\delta}$). Event-time dummies capture the first-giveaway effect; repeat_event$_{it}$ absorbs the repeat-giveaway level shift. The estimated $\hat{\delta} = 0.0001$ ($p = 0.999$) confirms that repeat giveaways add no incremental Steam engagement.

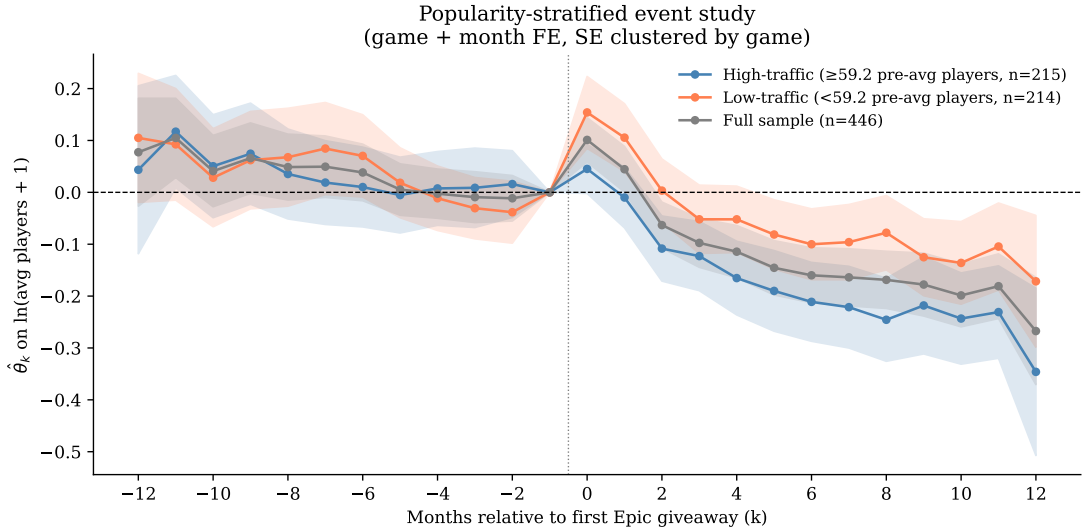

Figure 6: Event-study estimates by pre-treatment popularity. Split at the pre-treatment median (59.2 avg players/month): high-traffic ($N = 215$ games, left) and low-traffic ($N = 214$ games, right). High-traffic pre-trend Wald test: $\chi^2 = 9.35$, $p = 0.589$. Low-traffic: $\chi^2 = 19.31$, $p = 0.056$.

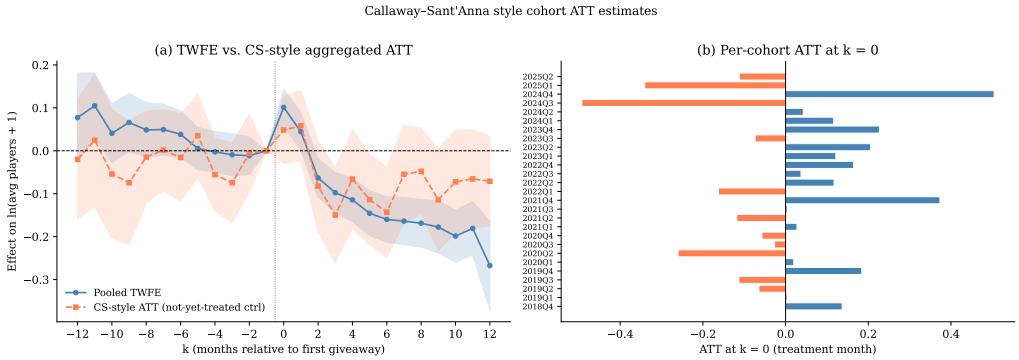

Figure 7: Callaway–Sant'Anna cohort-aggregated ATT estimates. Not-yet-treated games serve as controls within each treatment-quarter cohort; estimates are aggregated across 27 cohorts weighted by cohort size. Omitted period: $k = -1$.

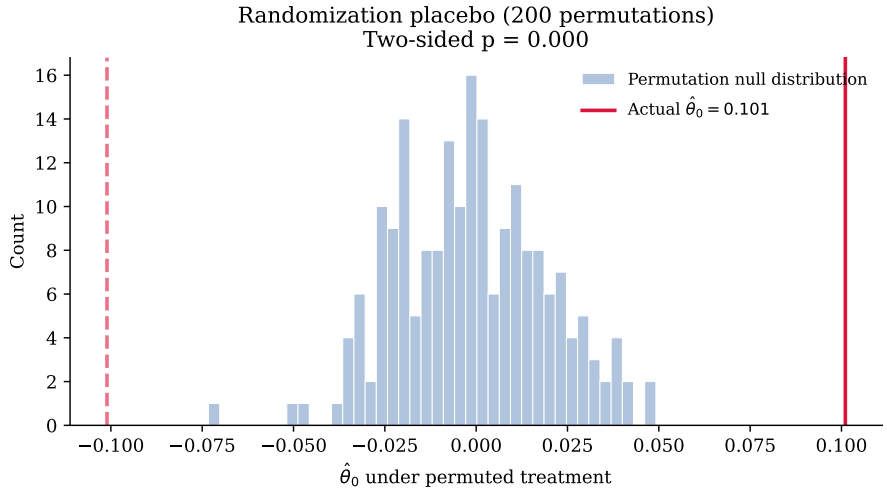

Figure 8: Randomization placebo test. Histogram of $|\hat{\theta}_0|$ from 200 permutations of treatment dates (preserving the empirical distribution of treatment months). Red vertical line: actual estimate $|\hat{\theta}_0| = 0.101$. The 95th percentile of the permutation null is 0.038; empirical $p < 0.001$.

Appendix

A. Clean-Sample Robustness Check

Figure 9 presents the pooled TWFE event-study on the clean sample (dropping all 30 games with a proximate repeat giveaway within 12 months), leaving 416 single-event games. Coefficients are nearly identical to the full-sample pooled estimates, confirming that the first-giveaway estimates are not driven by repeat-contamination bias.

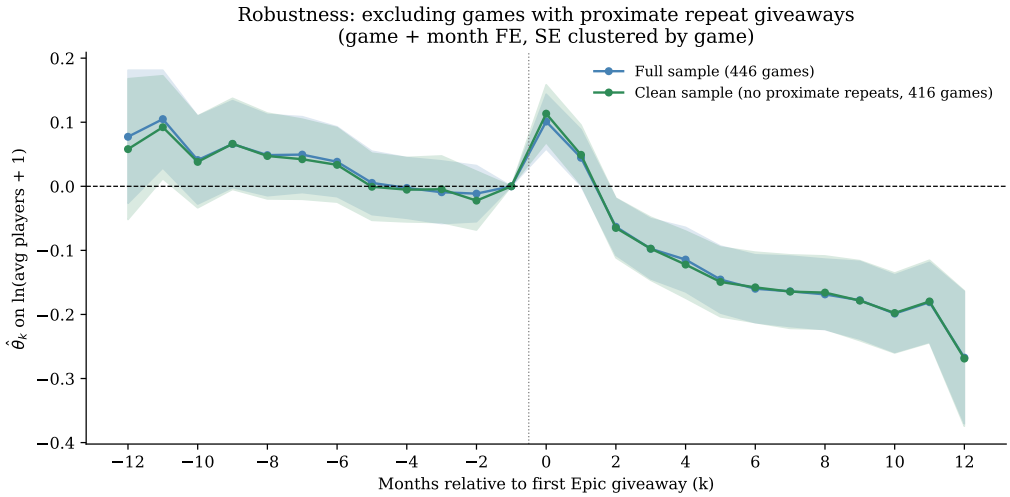

Figure 9: Clean-sample robustness check. Pooled TWFE re-estimated after dropping all 30 games with a proximate repeat giveaway (within 12 months), leaving 416 games. Coefficients are virtually identical to Figure 3.

B. Per-Cohort Event-Study Paths

Figure 10 overlays the full event-study path $\widehat{\mathrm{ATT}}(g,k)$ for each treatment cohort, coloured from early (dark purple) to late (bright yellow) by quarter of first-giveaway exposure. Lines are semi-transparent to reveal the density of trajectories; the thick black dashed line reproduces the weighted CS aggregate from Figure 7(a). Despite heterogeneity across cohorts, the common pattern of a sharp positive spike at $k = 0$ followed by gradual mean-reversion is visible throughout, corroborating the aggregate estimate.

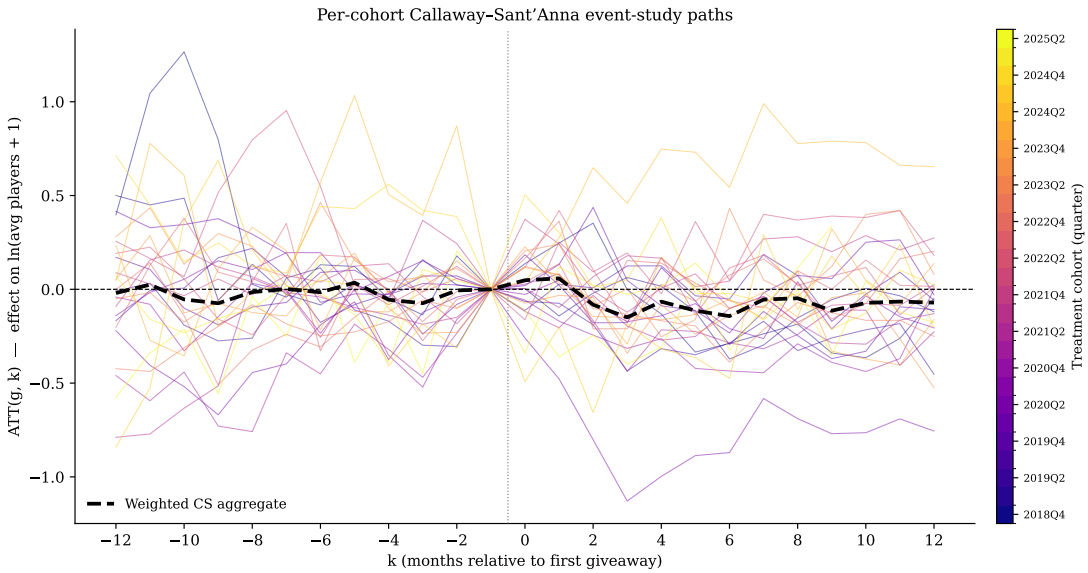

Figure 10: Per-cohort Callaway–Sant'Anna event-study paths. Each thin line traces $\widehat{\mathrm{ATT}}(g,k)$ for one treatment cohort g across event times $k \in [-12, 12]$; cohorts are coloured chronologically (early $\rightarrow$ late: purple $\rightarrow$ yellow, *plasma* palette). The thick black dashed line is the cohort-size-weighted CS aggregate. Vertical dotted line marks $k = -0.5$; horizontal dashed line marks zero.

C. Full TWFE Coefficient Table

Table 6 reports all 24 event-time coefficients from the pooled TWFE specification (Equation (1)), including confidence intervals.

Table 6: Full pooled TWFE event-study coefficients ($\hat{\theta}_k$)

k	Estimate	SE	p-value	2.5%	97.5%
−12	0.077	0.053	0.145	−0.027	0.181
−11	0.105	0.039	0.007	0.028	0.181
−10	0.041	0.035	0.242	−0.028	0.110
−9	0.066	0.035	0.057	−0.002	0.134
−8	0.049	0.032	0.132	−0.015	0.112
−7	0.049	0.030	0.102	−0.010	0.109
−6	0.038	0.028	0.168	−0.016	0.093
−5	0.005	0.025	0.833	−0.044	0.055
−4	−0.003	0.024	0.915	−0.050	0.045
−3	−0.009	0.025	0.710	−0.058	0.040
−2	−0.011	0.022	0.608	−0.055	0.032
−1	0 (ref)	—	—	—	—
0	**0.101*****	0.022	<0.001	0.058	0.144
1	0.045*	0.022	0.047	0.001	0.089
2	−0.063**	0.023	0.005	−0.108	−0.019
3	−0.097***	0.024	<0.001	−0.145	−0.050
4	−0.114***	0.026	<0.001	−0.165	−0.064
5	−0.145***	0.027	<0.001	−0.198	−0.093
6	−0.160***	0.027	<0.001	−0.213	−0.107
7	−0.164***	0.028	<0.001	−0.219	−0.109
8	−0.169***	0.028	<0.001	−0.224	−0.114
9	−0.178***	0.031	<0.001	−0.238	−0.117
10	−0.199***	0.031	<0.001	−0.259	−0.138
11	−0.181***	0.032	<0.001	−0.243	−0.119
12	−0.267***	0.053	<0.001	−0.371	−0.164

Notes: TWFE, $N = 38{,}641$; game and calendar-month fixed effects; standard errors clustered by game. $R^2 = 0.884$; within $R^2 = 0.008$.

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Note on AI usage: I used AI to assist with drafting and formatting this document in LaTeX.